Chapter 24 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

How is speciation in an isolated population connected to a change in the environment? **(BIG IDEA 1)**

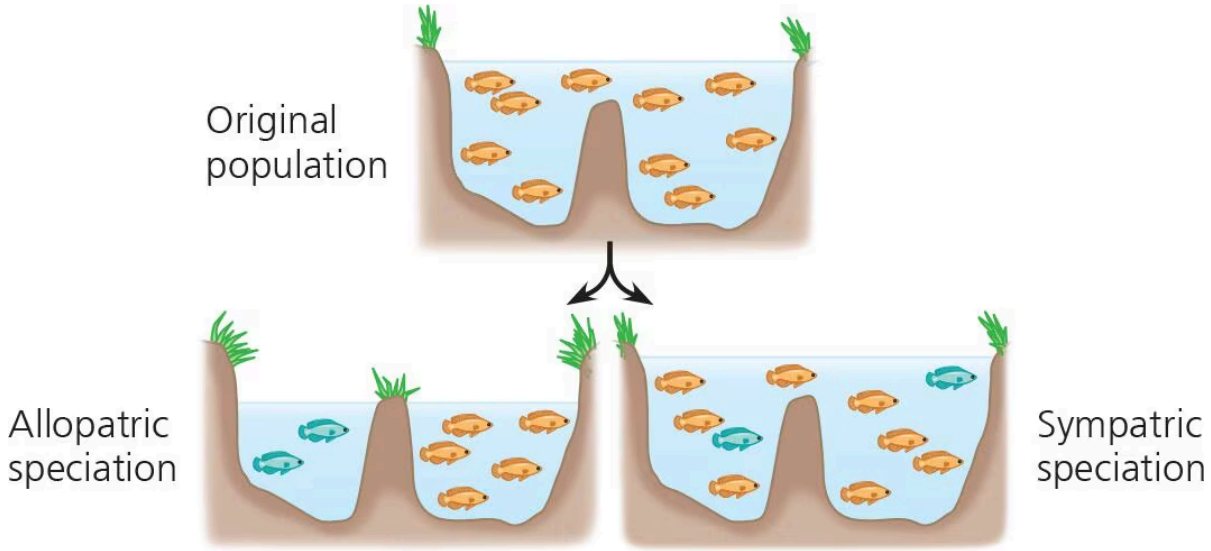
Concept 24.1: The biological species concept emphasizes reproductive isolation

- A biological **species** is a group of populations whose individuals may interbreed and produce viable, fertile offspring with each other but not with members of other species.
- New species form when **reproductive isolation** between populations develops through the establishment of **prezygotic** or **postzygotic barriers** that separate gene pools.

Explain the role of gene flow in the biological species concept.

Concept 24.2: Speciation can take place with or without geographic separation

- In **allopatric speciation**, gene flow is reduced when two populations of one species become geographically separated from each other. One or both populations may undergo evolutionary change during the period of separation, resulting in the establishment of barriers to reproduction.



A pool of gold fish are shown with a barrier in the middle that is partially submerged. If the water level drops, the pools are divided and allopatric speciation can occur which is shown as two blue fish in the left pool and five gold fish in the right pool. If the water level remains unchanged sympatric speciation can occur which is shown as two blue fish among eight gold fish.

- In **sympatric speciation**, a new species originates while remaining in the same geographic area as the parent species. Plant species (and, more rarely, animal species) have evolved sympatrically through **polyploidy**. Sympatric speciation can also result from sexual selection and habitat shifts.

Can factors that cause sympatric speciation also cause allopatric speciation? Explain.

Concept 24.3: Hybrid zones reveal factors that cause reproductive isolation

- Many groups of organisms form **hybrid zones** in which members of different species meet and mate, producing at least some offspring of mixed ancestry.
- Many hybrid zones are *stable*, in that hybrid offspring continue to be produced over time. In others, **reinforcement** strengthens prezygotic barriers to reproduction, thus decreasing the formation of unfit hybrids. In still other hybrid zones, barriers to reproduction may weaken over time, resulting in the *fusion* of the species' gene pools (reversing the speciation process).

What factors can support the long-term stability of a hybrid zone if the parent species live in different environments?

Concept 24.4: Speciation can occur rapidly or slowly and can result from changes in few or many genes

- New species can form rapidly once divergence begins, but it can take millions of years for that to happen. The time interval between speciation events varies considerably, from a few thousand years to tens of millions of years.
- Researchers have identified particular genes involved in some cases of speciation. Speciation can be driven by a few or many genes.

Is speciation something that happened only in the distant past, or are new species continuing to arise today? Explain.